ARJUN TYAGI

Urbana, IL · https://www.linkedin.com/in/arjun252001/ · arjun25@illinois.edu · 217-721-2414

SUMMARY

- Ph.D. student in Computer Architecture, specializing in cross-stack development and integration of processing-using-memory systems
- Extensive experience in memory systems, from devices to circuits, gained through multiple research internships

EDUCATION

University of Illinois Urbana-Champaign — **Champaign, IL**
Ph.D in Computer Science — *Expected May 2029*

Birla Institute of Technology & Science, Pilani — **Goa, India**
Bachelor of Engineering in Electronics and Instrumentation Engineering — *Aug. 2019 - May 2023*

PUBLICATIONS & WORKSHOPS

A. Tyagi and S. Sahay, *Efficient Reinforcement Learning On Passive RRAM Crossbar Array*, Submitted to IEEE Transactions on Very Large Scale Integration Systems (TVLSI), preprint available upon request

A. Tyagi and S. Kvatinsky, *Assessing the Performance of Stateful Logic in 1-Selector-1-RRAM Crossbar Arrays*, 2024 IEEE International Symposium on Circuits and Systems (ISCAS), pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558539

ARCANA Research Group and Sandia National Laboratories, *Tutorial on Simulation for Processing-Using-Memory Systems*, 2024 IEEE International Symposium on Computer Architecture (ISCA) Workshop

RESEARCH EXPERIENCE

ARCANA Research Group, PI: Prof. Saugata Ghose — **Urbana, IL**

Rapid, Accurate Chip-Scale Simulation of Digital Compute Using Emerging Memory Devices — *Sept. 2023 - present*

- Developed a fast simulation framework for digital in-memory computing, enabling chip-scale modeling with $> 10,000\times$ speedup over SPICE tools while maintaining high accuracy
- Introduced techniques such as memoization and dynamic time stepping to drastically reduce simulation time and memory usage while still being robust to device variability
- Enabled seamless integration of Verilog-A device models, supporting iterative co-design across device, circuit, and architecture levels
- Paper in preparation for submission to DAC 2026

ASIC^2 Research Group, PI: Prof. Shahar Kvatinsky — **Haifa, Israel**

Assessing the Performance of Stateful Logic in 1-Selector-1-RRAM Crossbar Arrays — *Jan. - Jun. 2023*

- Developed a compact Verilog-A model for 1S1R memory cells, aimed for large-scale simulations
- Modeled parasitic RC effects for intra-array interconnects to realistically evaluate the effect of array size on performance degradation
- Simulated passive (1R) and active (1S1R) RRAM arrays of varying sizes (4×4 to 512×512) for in-memory computing using MAGIC logic gates
- Demonstrated that 1S1R arrays offer up to $24.4\times$ lower power and $4.5\times$ higher readout margin than 1R arrays, enabling reliable scaling to large arrays

NeuroChaSe Group, PI: Prof. Shubham Sahay **Kanpur, India**

Efficient Reinforcement Learning on Passive RRAM Crossbar Array *Jan. - Jul. 2022*

- Performed an algorithm-hardware co-design and proposed a novel implementation of Monte Carlo reinforcement learning algorithm on passive RRAM crossbar array
- Demonstrated $> 100,000\times$ area reduction over prior digital and 1T1R-based implementations, while maintaining robustness to spatial/temporal variations and endurance failures in RRAM devices

SKILLS & ABILITIES

- **Languages/Applications**: C, C++, Python, Verilog HDL, Verilog A, MATLAB
- **Graduate Courses**: Programming Languages, Compiler Construction, Computer System Organization, Parallel Computer Architectures, Emerging Memory & Storage Systems
- **Undergraduate Courses**: Microelectronic Circuits, Analog and Digital VLSI Design, Analog Electronics, Digital Design

TUTORING & VOLUNTEERING EXPERIENCE

Graduate Student Ambassador at UIUC CS *Spring'25*

Undergraduate/Graduate Mentor, UIUC

· Dylan Meeks, ECE '26 at University of Texas at Austin *Jun. 2025 - Aug. 2025*